

Data Flow Diagram

Date	29 SEPTEMBER 2026
Team ID	SWTID-2026-6684
Project Name	ComicCraft-AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	User (Browser Client), External AI APIs (Google Gemini API, Hugging Face Diffusers API).
Rectangle with numbered header	Process	• 1.0 Process User Input & Router (routes.py / FastAPI) • 2.0 Generate Outline & Story (gemini_flash.py, gemini_pro.py) • 3.0 Generate Illustrations (image_generator.py using Stable Diffusion) • 4.0 Build Layout & Export PDF (layout_builder.py, exporters.py)
Rectangle (solid fill, no number)	Data Store	• D1 Static Panel Images (static/panels/)

		• D2 Exported PDFs (<code>static/exports/</code>)
Labeled arrow	Data Flow	• User $\rightarrow$ Process 1.0: Submits story parameters (Prompt, Character, Setting, Tone, Style) via <code>index.html</code> or <code>/generate-comic/json</code>. • Process 1.0 $\rightarrow$ Process 2.0: Sends combined prompt details to Gemini API integrations. • Process 2.0 $\rightarrow$ Gemini API $\rightarrow$ Process 2.0: Receives structured 5-panel JSON outline (Gemini Flash) and full narration/dialogue text (Gemini Pro). • Process 2.0 $\rightarrow$ Process 3.0: Sends image prompts extracted from outline to Stable Diffusion pipeline. • Process 3.0 $\rightarrow$ Data Store D1: Saves generated comic images to <code>static/panels/</code>. • Process 3.0 & Process 2.0 $\rightarrow$ Process 4.0: Passes image paths, narration text, and panel metadata to layout builder. • Process 4.0 $\rightarrow$ Data Store D2: Compiles PDF via FPDF and saves to <code>static/exports/</code>. • Process 4.0 $\rightarrow$ User: Renders preview on <code>comic_preview.html</code> and provides downloadable PDF link.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

